(§) Storing the array in reverse order, deleting last element might just be a matter of shrinking the size of the array, or somehow keep track of what's the last element in the array.

Moreover, consider that most of the times containers, and priority queues in particular, are used as support structures, meaning that they are part of more complex algorithms/data structures and each cycle of the main algorithm can call operations on the PQ several times: for example for a sorting algorithm, this could mean going from `O(n2)` (which usually means unfeasible for `n` as large as 1 million, or even less) to `O(n*log(n))`, which is still tractable for inputs of size 1 billion or more[6].

However, this would come at a cost, since the implementation of balanced binary trees is usually not trivial.

Next, we present a way to efficiently implement a generic priority queue.

2.5.3 Heap

A binary heap is the most used version of priority queues. It allows elements to be inserted and retrieved in either ascending or descending order, one at the time.

While in reality an underlying array is used to store heap's elements, it can conceptually be represented as a particular kind of binary tree, abiding by 3 invariants:

1. Every node has at most 2 children.